Angles & Parallel Lines Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Vertical angles

ORIGINAL PROBLEM

Two lines intersect. One angle is $(3x + 10)°$ and its vertical angle is $(5x − 20)°$. Find x.

INCORRECT WORKED SOLUTION

$(3x + 10) + (5x − 20) = 180$

$8x − 10 = 180$

$8x = 190$

$x = \frac{95}{4}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Linear pair

ORIGINAL PROBLEM

Adjacent angles on a straight line measure $(2x + 20)°$ and $(x + 40)°$. Find x.

INCORRECT WORKED SOLUTION

$2x + 20 = x + 40$

$x = 20$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Corresponding angles

ORIGINAL PROBLEM

a ∥ b. Corresponding angles measure (3x + 5)° and (5x − 25)°. Find x.

INCORRECT WORKED SOLUTION

(3x + 5) + (5x − 25) = 180

8x − 20 = 180

8x = 200

x = 25

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Same-side interior angles

ORIGINAL PROBLEM

a ∥ b. Same-side interior angles measure (3x + 15)° and (2x + 10)°. Find x.

INCORRECT WORKED SOLUTION

3x + 15 = 2x + 10

x = −5

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Proving lines parallel

ORIGINAL PROBLEM

Given: ∠1 ≅ ∠2 (alternate interior angles).
Prove: a ∥ b.

INCORRECT WORKED SOLUTION

1. a ∥ b
2. Alternate interior angles are congruent.
3. ∠1 ≅ ∠2

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Angle bisector

ORIGINAL PROBLEM

BD bisects ∠ABC. m∠ABC = 100° and m∠ABD = (4x + 10)°. Find x.

INCORRECT WORKED SOLUTION

4x + 10 = 100

4x = 90

x = 22.5

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Complement word problem

ORIGINAL PROBLEM

An angle measures 20° more than its complement. Find the angle.

INCORRECT WORKED SOLUTION

Let x = the complement. The angle is x + 20.

x + (x + 20) = 180

2x = 160

x = 80, so the angle is 100°.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Solving carefully

ORIGINAL PROBLEM

a ∥ b. Alternate interior angles measure (7x − 9)° and (4x + 21)°. Find x and the angle measure.

INCORRECT WORKED SOLUTION

7x − 9 = 4x + 21

3x − 9 = 21

3x = 12

x = 4

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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9. Angles around a point

ORIGINAL PROBLEM

Three angles around a point measure (2x)°, (3x + 10)° and (x + 50)°. Find x.

INCORRECT WORKED SOLUTION

2x + 3x + 10 + x + 50 = 180

6x + 60 = 180

6x = 120

x = 20

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Naming angle pairs

ORIGINAL PROBLEM

a ∥ b with transversal t. ∠4 and ∠5 are between the lines on the same side of t. m∠4 = 112°. Find m∠5.

INCORRECT WORKED SOLUTION

∠4 and ∠5 are alternate interior angles.

Alternate interior angles are congruent.

m∠5 = 112°

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points